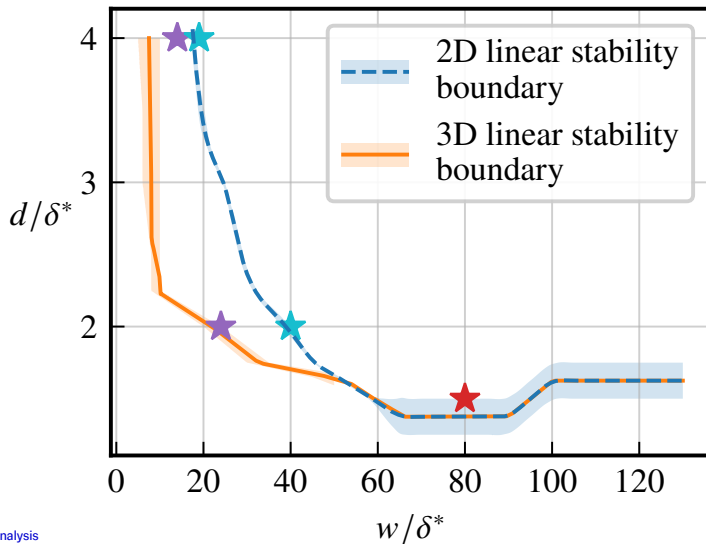


IMPERIAL

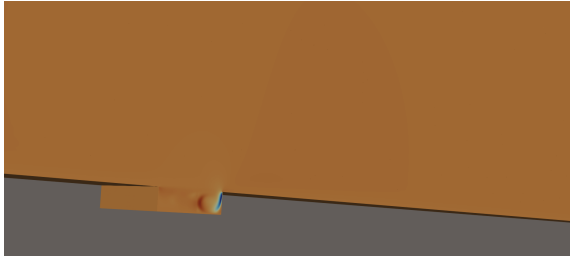
Re analysis

Víctor Ballester
July 23, 2026

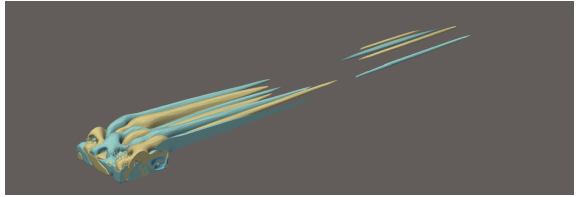
Summary of results (Re = 1000)



Case ★

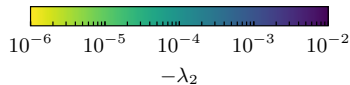
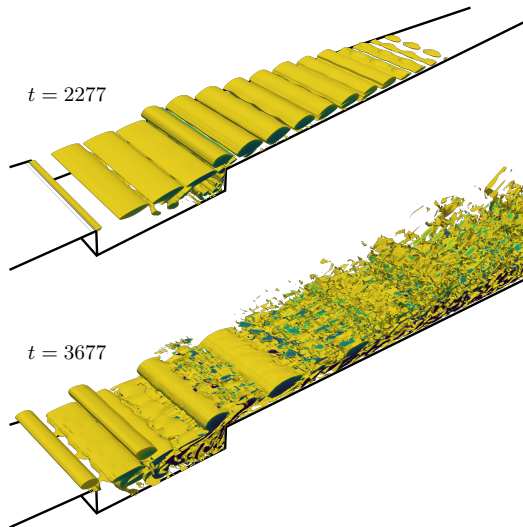
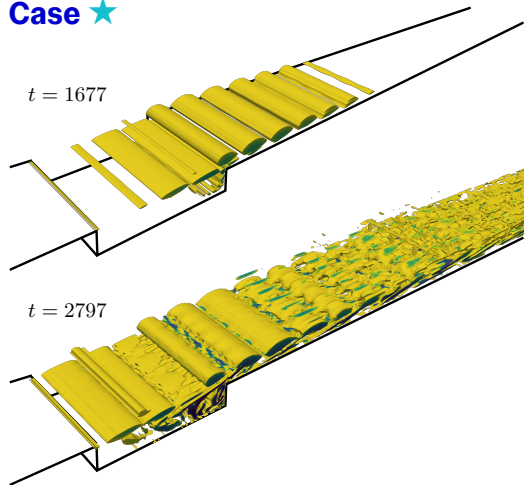


Full velocity v field (very similar to the base state)



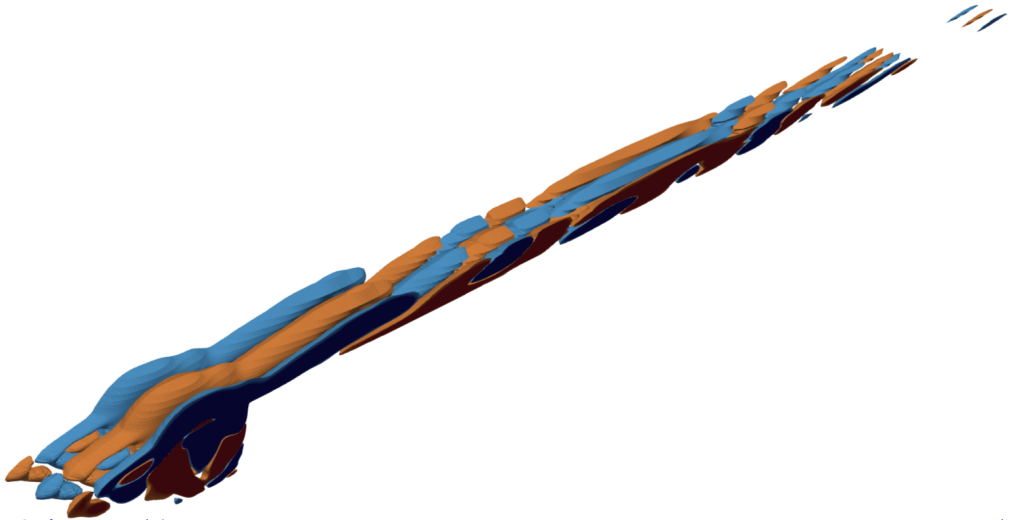
Perturbation v field after being evolved with the nonlinear operator

Case ★



Transition through secondary instability
(floquet analysis)

Case ★ - floquet mode



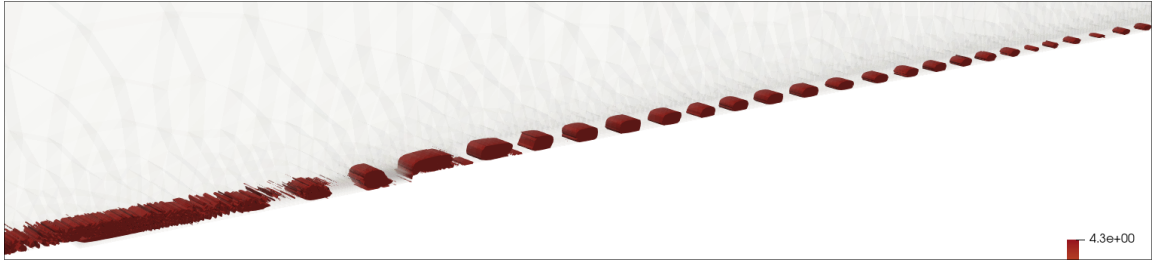
Case ★

About the appearance of the 2d mode first. Two diagnostics that I am currently doing:

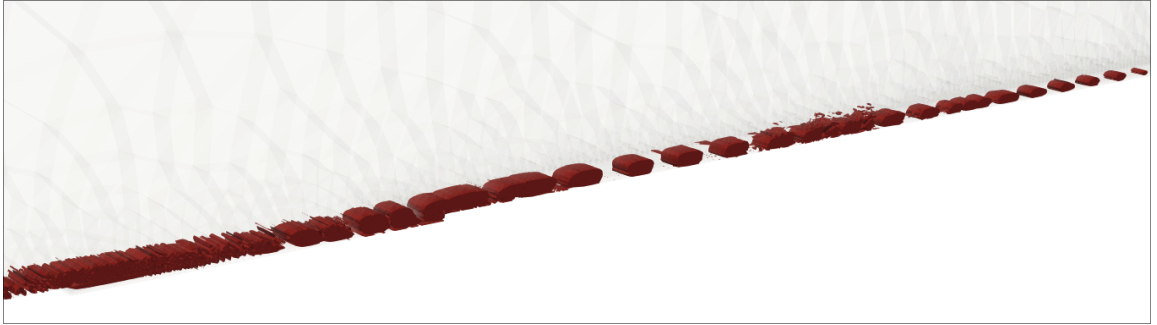
- First is run the direct solver with the 3d perturbation and the full set of modes (instead of just only one) to see if that modifies the growth rate of the whole 3d mode.
- I have the feeling that any small noise trigger quickly the 2d mode, and I coarsen the mesh (maybe slightly to much) and that is why the 2d mode might appear first.

Transition in case ★

Frequency of the most unstable 2d mode for this cavity is $\omega \approx 0.099$, which lies inside the unstable region of the neutral stability curve.



Transition in case ★



Transition in case ★

